

Aggregation in SQL – ASSIGNMENT

Q1. Count cities in each country

SQL: SELECT c.Name, COUNT(ci.ID) FROM Country c LEFT JOIN City ci ON c.Code = ci.CountryCode GROUP BY c.Name;

Explanation: Counts number of cities per country.

Q2. Continents with more than 30 countries

SQL: SELECT Continent, COUNT(*) FROM Country GROUP BY Continent HAVING COUNT(*) > 30;

Explanation: HAVING filters aggregated results.

Q3. Regions with population over 200 million

SQL: SELECT Region, SUM(Population) FROM Country GROUP BY Region HAVING SUM(Population) > 200000000;

Explanation: SUM calculates total population.

Q4. Top 5 continents by average GNP

SQL: SELECT Continent, AVG(GNP) FROM Country GROUP BY Continent ORDER BY AVG(GNP) DESC LIMIT 5;

Explanation: Shows economically strongest continents.

Q5. Official languages per continent

SQL: SELECT c.Continent, COUNT(cl.Language) FROM Country c JOIN CountryLanguage cl ON c.Code = cl.CountryCode WHERE cl.IsOfficial = 'T' GROUP BY c.Continent;

Explanation: Counts only official languages.

Q6. Max and Min GNP per continent

SQL: SELECT Continent, MAX(GNP), MIN(GNP) FROM Country GROUP BY Continent;

Explanation: Finds richest and poorest countries.

Q7. Country with highest average city population

SQL: SELECT c.Name, AVG(ci.Population) FROM Country c JOIN City ci ON c.Code = ci.CountryCode GROUP BY c.Name ORDER BY AVG(ci.Population) DESC LIMIT 1;

Explanation: Uses AVG and ORDER BY.

Q8. Continents with avg city population > 200000

SQL: SELECT c.Continent, AVG(ci.Population) FROM Country c JOIN City ci ON c.Code = ci.CountryCode GROUP BY c.Continent HAVING AVG(ci.Population) > 200000;

Explanation: HAVING works on aggregated data.

Q9. Population & life expectancy per continent

SQL: SELECT Continent, SUM(Population), AVG(LifeExpectancy) FROM Country GROUP BY Continent ORDER BY AVG(LifeExpectancy) DESC;

Explanation: Ordered by life expectancy.

Q10. Top 3 continents by life expectancy (Pop > 200M)

SQL: SELECT Continent, AVG(LifeExpectancy), SUM(Population) FROM Country GROUP BY Continent HAVING SUM(Population) > 200000000 ORDER BY AVG(LifeExpectancy) DESC LIMIT 3;

Explanation: Combines HAVING and LIMIT.